# ANALYTICAL CUMULATIVE EXAM
**October 2, 1993**

**The general themes of this cume are:**
**Acid/Base Equilibria and Isoelectric Focusing**

**(10 pts)** **1.** Alanine is an amino acid that can assume three different forms. The pH of the solution will determine which of the three forms will be in the highest concentration. The form of alanine that exists at intermediate pH values is shown below in equilibrium with its zwitterion.

(a) Draw the form that alanine assumes at low pH.
(b) Draw the form that alanine assumes at high pH.

**(16 pts)** **2.** (a) (3 pts) Which form of alanine will not migrate in an electric field?
(b) (3 pts) Which form of alanine will migrate toward the negative pole of an electric field?
(c) (3 pts) which form of alanine will migrate toward the positive pole of an electric field?
(d) (7 pts) In reality, there are acid/base equilibria set up among the three forms of alanine. If there is to be *no net* migration of alanine in an electric field, what must the concentration *ratio* be of [the form that migrates towards the negative pole] / [the form that migrates towards the positive pole]?

**(10 pts)** **3.** Write out the equilibrium expressions which describe the acid dissociation constants $K_1$ and $K_2$. Let $K_1$ be the acid dissociation constant of the diprotic form, and $K_2$ be the acid dissociation constant of monoprotic form.

**(40 pts)** **4.** The pH at which the average charge of a polyprotic acid is zero is called the *isoelectric point.* (This is also the pH at which the net migration of the polyprotic acid is zero within an electric field.) For a molecule such as alanine, this hydrogen ion concentration can be calculated from the following equation:

$$[H^+] = \sqrt{K_1 K_2}$$

Derive this equation. (Hint: use the answers from parts **3** and **2d.**)

**(15 pts)** **5.** The isoelectric point is the basis of a very sensitive technique of protein separation called isoelectric focusing. A mixture of proteins is placed at one end of a tube filled with polyacrylamide gel. A strong electric field is placed across the length of the gel. The gel is engineered so that it contains a stationary pH gradient from low to high pH, that extends from the positive pole to the negative pole of the electric field, respectively. The proteins migrate in one direction or the other until they reach the region of their isoelectric pH. In this region they have no net charge and no longer move. Thus each protein in the mixture is focused in one small region (of the gel) at its isoelectric pH.

A linear pH gradient from 0 to 10 pH is established along a 10 cm-long gel. At what distance from the low-pH-end of the gel would you expect to see a band of the following compounds after isoelectric focusing?

(a) alanine, $pK_1 = 2.35$, $pK_2 = 9.87$

(b) serine, $H_3N^+CH(CH_2OH)COO^-$, $pK_1 = 2.2$, $pK_2 = 9.1$

(c) ethanol

**(9 pts)** **6.** What do you expect to happen if a protein diffuses out of its isoelectric region during and isoelectric focusing experiment?